

IVACS V-TRACE: Vehicle Trust, Route & Evidence Engine

Problem Statement OD-08: License Plate Detection and Recognition from Construction-Site CCTV Footage

Report Updated: 2026-09-18 11:50:01	Current State: Demo Engine & Scenarios Verified	Runtime AI Policy: Strict Zero LLM / Native CV
Team: CodeAvengers	Evaluation Status: DEMO READY (22/22 Tests Pass)	Git Branch: main (Synced Remote)

1. System Component Status & Subsystem Audit

Component / Subsystem	Location / Path	Operational Description	Status
Video Ingestion Engine	backend/video/	MP4, RTSP, Webcam stream abstraction with configurable frame stride	PASS
Vehicle Detection Engine	backend/detection/vehicle_detector.py	Ultralytics YOLOv8n inference filtering cars, trucks, buses, motorcycles	PASS
Plate Localization Engine	backend/detection/plate_detector.py	Bumper ROI localization, morphological gradients, and vehicle bbox association	PASS
OCR Character Engine	backend/ocr/plate_ocr.py	EasyOCR recognition with IND blue strip removal and slot normalization	PASS
Vehicle Visual Fingerprint	backend/vehicle_identity/feature_extractor.py	ResNet18 backbone yielding 512-dim L2-normalized deep visual embeddings	PASS
Identity Decision Engine	backend/vehicle_identity/identity_rules.py	Deterministic Rules A-E for plate-visual match, mismatch, and swap detection	PASS
Demo Scenario State Machine	backend/demo/scenario_manager.py	Controlled 4-scenario runner: Normal Repeat, Identity Mismatch, Plate Swap, Unreadable Plate	PASS
Evidence Comparison System	backend/app.py	Side-by-side evidence image endpoint and observation inspection with fallback	PASS
Temporal Deduplication Engine	backend/vehicle_identity/identity_service.py	5-second cooldown suppression per (vehicle_id, plate) preventing duplicate event floods	PASS
Database Persistence Layer	backend/database/	SQLite persistence for detections, registered identities, demo-flagged observations, and isolated reset	PASS
Evidence Storage Engine	data/evidence/	Multi-scale image storage (full frame, vehicle crop, plate crop, annotated ROI)	PASS
REST API Application	backend/app.py	FastAPI web service with video streaming, scenario execution, and reset endpoints	PASS
Automated Test Suite	tests/	22 comprehensive unit, integration, and scenario tests covering all pipeline modules	PASS

2. Team Member Role Allocation (CodeAvengers Team)

Member	GitHub / Email	Subsystem Ownership & Core Deliverables
Jone Kavish (Team Lead)	jonekavish-dot jonekavish@gmail.com	Backend Architecture & Evidence System: FastAPI service, evidence comparison endpoints, EasyOCR pipeline, ResNet18 visual fingerprint engine, database schema & isolated demo reset handler, project leadership.
K.V. Pranesh (Member 1)	kvpranesh kvpranesh49@gmail.com	Vehicle Detection & Temporal Deduplication: Ultralytics YOLOv8n detector, COCO vehicle filtering, bounding box tracking, 5-second temporal cooldown deduplication on identity events.
Gowshik Gunal (Member 2)	gowshikgunal22 gowshikgunal@gmail.com	Plate Localization & Demo State Machine: Bumper ROI localization, morphological gradients, 4-scenario demo state machine implementation, step-by-step frame injection.

Dinesh Balu (Member 3)	dineshbalu7f-glitch dineshbalu7.f@gmail.com	Persistence, Media Generator & Test Suite: Video source abstraction, offline deterministic scenario media generator, 8-test scenario QA suite, SQLite persistence layer.
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3. Vehicle Identity Rules & Controlled Hackathon Demo Scenarios

IVACS V-TRACE implements deterministic, calibrated comparison between the current vehicle's 512-dimensional ResNet18 feature embedding and historical records. In strict accordance with hackathon evaluation guidelines, the system employs an **Honest Detection Policy**: it never claims unverified ground truth (e.g., 'stolen' or 'cloned'), emitting qualified analytical signals instead: POSSIBLE PLATE-VEHICLE IDENTITY MISMATCH and MANUAL VERIFICATION REQUIRED.

Rule	Condition / Observation	Analytical Event Signal	Operational Meaning
Rule A	Same Plate + High Similarity (≥ 0.85)	SAME_VEHICLE	Confirmed consistent identity. Observation appended to history.
Rule B	Same Plate + Low Similarity (< 0.85)	POSSIBLE_IDENTITY_MISMATCH	Visual appearance deviates from registered record for this plate. Possible plate cloning.
Rule C	Different Plate + High Similarity (≥ 0.85)	POSSIBLE_PLATE_SWAP	High visual match with a previously seen vehicle bearing a different plate. Possible plate swap anomaly.
Rule D	Plate Unreadable + High Similarity (≥ 0.85)	PLATE_UNREADABLE_VEHICLE_MATCH	Plate obscured or blurry, but vehicle visual fingerprint matches existing identity. Tracking continuity preserved.
Rule E	No Historical Match Found	NEW_VEHICLE	First sighting of this vehicle. A new persistent identity record is registered in the database.

Controlled Hackathon Demo Scenarios:

Scenario ID	Step 1 (Baseline)	Step 2 (Trigger)	Expected Decision & Similarity
NORMAL_REPEAT	White Sedan (MH12DE1433)	White Sedan (MH12DE1433)	SAME_VEHICLE (sim ≥ 0.90)
IDENTITY_MISMATCH	White Sedan (MH12DE1433)	Red Truck (MH12DE1433)	POSSIBLE_IDENTITY_MISMATCH (sim ≤ 0.35)
PLATE_SWAP	White Sedan (MH12DE1433)	White Sedan (KA01AB1234)	POSSIBLE_PLATE_SWAP (sim ≥ 0.88)
PLATE_UNREADABLE	White Sedan (MH12DE1433)	White Sedan (NO_PLATE/BLUR)	PLATE_UNREADABLE_VEHICLE_MATCH (sim ≥ 0.88)

4. REST API Endpoint Specifications

Method	Endpoint Route	Description & Operational Response
GET	/api/health	Returns system health, active compute device (CPU/CUDA), and model status.
GET	/api/cameras	Lists all configured construction-site CCTV camera feeds (CAM-01 through CAM-04).
POST	/api/demo/scenario/start	Starts one of 4 controlled scenarios (NORMAL_REPEAT, IDENTITY_MISMATCH, PLATE_SWAP, PLATE_UNREADABLE).
POST	/api/demo/scenario/stop	Stops active scenario runner gracefully.
GET	/api/demo/scenario/status	Real-time scenario state: active scenario, step index, decision, similarity, crops.
POST	/api/demo/reset	Clears all demo records (is_demo=1) while strictly preserving system tables and production data.
GET	/api/vehicles/{id}/comparison	Returns side-by-side evidence: current vehicle/plate vs historical vehicle/plate, similarity, alert text.
GET	/api/identity-events/{id}	Fetches single identity observation by ID including similarity score and evidence crop paths.
GET	/api/detections	Paginated list of structured vehicle & plate detection events stored in SQLite.
GET	/api/vehicles	Catalog of all registered vehicle identities with visit counts and canonical plates.
GET	/api/identity-events	Chronological stream of vehicle identity matching decisions and similarity scores.

GET	/evidence/{filename}	Static HTTP asset access to saved evidence frames, vehicle crops, and plate crops.
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5. Automated Quality Assurance & Test Verification (22/22 Passed)

Automated test suite executed via python -m pytest tests/ -v. Pass rate: 100% (22 passed, 0 failed) in 29.17s.

Module	Test Name	Verification Scope	Result
tests/test_demo_scenarios.py	test_scenario_normal_repeat	Verifies Rule A normal repeat execution and SAME_VEHICLE event	PASS
tests/test_demo_scenarios.py	test_scenario_identity_mismatch	Verifies Rule B mismatch: same plate + different appearance yields sim < 0.35	PASS
tests/test_demo_scenarios.py	test_scenario_plate_swap	Verifies Rule C plate swap: different plate + same appearance yields sim >= 0.85	PASS
tests/test_demo_scenarios.py	test_scenario_unreadable_plate	Verifies Rule D unreadable plate: tracking continuity maintained via embedding	PASS
tests/test_demo_scenarios.py	test_scenario_start_stop	Validates non-blocking scenario manager start and graceful stop transitions	PASS
tests/test_demo_scenarios.py	test_scenario_reset	Ensures POST /api/demo/reset purges only demo records without schema alteration	PASS
tests/test_demo_scenarios.py	test_comparison_api	Validates GET /api/vehicles/{id}/comparison returns side-by-side evidence paths	PASS
tests/test_demo_scenarios.py	test_duplicate_identity_event_suppression	Asserts 5-second temporal cooldown deduplication suppresses rapid event bursts	PASS
tests/test_identity.py	test_cosine_similarity_basics	Verifies vector math, clamping [-1.0, 1.0], and orthogonal handling	PASS
tests/test_identity.py	test_rule_new_vehicle	Validates Rule E: Registration of new vehicle identity on first sighting	PASS
tests/test_identity.py	test_rule_same_vehicle	Validates Rule A: High similarity + same plate yields SAME_VEHICLE	PASS
tests/test_identity.py	test_rule_possible_identity_mismatch	Validates Rule B: Same plate + low similarity yields POSSIBLE_IDENTITY_MISMATCH	PASS
tests/test_identity.py	test_rule_possible_plate_swap	Validates Rule C: Different plate + high similarity yields POSSIBLE_PLATE_SWAP	PASS
tests/test_identity.py	test_rule_plate_unreadable_vehicle_match	Validates Rule D: Unreadable plate + high similarity yields tracking continuity	PASS
tests/test_identity.py	test_service_identity_lifecycle	End-to-end service test: registers identity, observation, and DB persistence	PASS
tests/test_vtrace.py	test_video_source_initialization	Verifies MP4Source open, metadata retrieval, frame count, and FPS	PASS
tests/test_vtrace.py	test_invalid_video_handling	Tests robust failure handling for non-existent video files	PASS
tests/test_vtrace.py	test_frame_processing_with_no_detection	Asserts blank frames produce None without throwing runtime exceptions	PASS
tests/test_ocr_empty_result	test_ocr_empty_result	Ensures unreadable plate crops handle gracefully with confidence 0.0	PASS
tests/test_vtrace.py	test_valid_structured_detection_event	Validates Pydantic DetectionEvent schema and slot normalization	PASS
tests/test_vtrace.py	test_database_insertion	Verifies SQLite record insertion and query retrieval of detection events	PASS
tests/test_vtrace.py	test_evidence_image_creation	Confirms evidence images and crops are saved with correct file paths	PASS

6. Git Traceability & Remote Commit Provenance

Remote Repository: https://github.com/jonekavish-dot/VISIONX_CODEAVENGERS_OD-08.git

All commits are authored team-member-wise and synchronized with the remote main branch.

Commit	Author	Date	Commit Message & Deliverable Scope
a16c6aa	K.V. Pranesh kvpranesh49@gmail.com	2026-09-18	feat(identity): add temporal cooldown deduplication and comparison schema definitions
997e488	Gowshik Gunal gowshikgunal@gmail.com	2026-09-18	feat(demo): implement 4-scenario deterministic demo state machine and evidence pipeline
6500f13	Dinesh Balu dineshbalu7.f@gmail.com	2026-09-18	feat(demo): build offline deterministic scenario media generator and 8-test validation suite
2bf6a26	Jone Kavish jonekavish@gmail.com	2026-09-18	docs(report): update progress report PDF to 3-page comprehensive executive audit with test matrix and git provenance
a68f7b3	Jone Kavish jonekavish@gmail.com	2026-09-18	feat(identity): implement Vehicle Visual Fingerprint engine, ResNet18 embeddings, decision rules A-E, and vehicle endpoints
339c7f1	Jone Kavish jonekavish@gmail.com	2026-09-18	fix(runner): add run_server.py entrypoint and update python -m uvicorn instructions
1e644db	Jone Kavish jonekavish@gmail.com	2026-09-18	feat(core, api): deliver IVACS V-TRACE foundation pipeline, EasyOCR engine, FastAPI server, tests and docs
5bab7cf	Gowshik Gunal gowshikgunal@gmail.com	2026-09-18	feat(plate-detect): implement bumper ROI license plate localization and vehicle association